



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

PROJECT TITLE: Hospital Management System for Radium Diagnostic & Consultation Center

Semester: Summer 25-26

<i>Group: C</i>		<i>Section: DD</i>
SL	Student Name	Student ID
1	SAMIA AFRUZ ASHA	24-57314-2
2	MUNISH SARKER	24-57375-2
3	RINKO RANI VADRA	24-58043-2
4		
5		

Assessment Rubric

CO1: Explain the impact of software engineering over various contexts of software development to assess societal, health, safety, legal and cultural issues.

Student ID	Problem Analysis [3]	Socio-cultural Impact [3]	Related Solutions and Studies [4]
1.			
2.			
3.			
4.			
5.			

CO2: Explain software engineering principles considering professional responsibilities and societal impacts.

Student ID	Conceptual Understanding [3]	Principle Selection and Justification [3]	Application in Software Development Context [4]
1.			
2.			
3.			
4.			
5.			

Table of Contents

1. Project Proposal	04-07
1.1 Background to the Problem	04-05
1.2 Selection of Process Model	05-07
2. Timeline Chart / Gantt Chart	08
3. Software Requirements Specification (SRS) / PRD	09-12
3.1 Scopes and Features	09
3.2 User Story Table	09-10
3.3 Requirements Traceability Matrix	11-12
3.3.1 Functional Requirements	11
3.3.2 Non-Functional Requirements	12
4. Software Design	13-32
4.1 System Design	13
4.2 UI Design using Figma	14-32
5. Conclusion	33
6. Appendix.....	33-34

1. PROJECT PROPOSAL

1.1 Background to the Problem

Healthcare organizations need an efficient system to manage their daily activities and provide better services to patients. Managing appointments, patient records, doctor schedules, diagnostic reports, and billing manually can take a lot of time and may lead to mistakes or delays. A Hospital Management System (HMS) can simplify these tasks by keeping all information in one place and making hospital operations more organized. Radium Diagnostic & Consultation Center, located in Daudkandi, Cumilla, offers specialist consultation and diagnostic services through different medical departments, including pathology, X-ray, ECG, and ultrasound. Its website provides useful information about doctors, departments, visiting schedules, and contact details. However, based on the publicly available information, patients still need to contact the hospital by phone to book appointments, and there is no publicly available system for online appointment booking, patient record management, or diagnostic reports. This gap means patients cannot check doctor availability or their own visit history without calling the center directly, and staff must coordinate appointments, records, and billing manually across separate registers. This project addresses that gap by proposing a web-based Hospital Management System that consolidates patient registration, appointment booking, doctor and department management, electronic medical records, diagnostic report handling, billing, and administrative reporting into a single, coherent platform, giving hospital staff and patients a shared, up-to-date view of each patient's care and account status.

These manual, phone-dependent processes create compounding difficulties across the diagnostic center's daily operations. Patients calling to book an appointment often face busy lines or unanswered calls, and once an appointment is confirmed, there is no automated reminder to reduce no-shows. Front-desk staff must cross-reference paper registers to check doctor availability, which slows down registration and increases the chance of double-booking. Diagnostic reports produced by pathology, X-ray, ECG, and ultrasound departments are typically handed to patients as physical printouts, so past results are not easily retrievable if a doctor needs to review a patient's history during a later visit, and files can be misplaced or damaged over time. Billing is calculated and recorded manually at each visit, which makes it difficult to track outstanding payments, generate accurate revenue reports, or reconcile accounts across departments. Because appointment, patient, and financial records are kept in separate paper systems, hospital administrators have no consolidated view of daily operations and must compile reports by hand when planning staffing or reviewing performance. As patient volume grows, these manual processes become harder to sustain, increasing the risk of lost records, scheduling conflicts, and delays in patient care.

To address these challenges, this project proposes designing and developing a web-based Hospital Management System (HMS) tailored to Radium Diagnostic & Consultation Center's operations. The system will replace the current phone-based, paper-driven workflow with a single online platform accessible to patients, doctors, receptionists, laboratory technicians, and administrators, each through a role-specific interface.

Patients will be able to register an account, search for doctors by department or specialty, view real-time appointment availability, and book or cancel appointments without calling the center; the system will also send automated reminders before scheduled visits and give patients access to their own visit history and diagnostic reports. Doctors will have access to a personal dashboard showing their upcoming schedule and the medical

records of patients they are consulting, reducing the time spent searching for paper files before an appointment. Receptionists will use the system to register walk-in patients, confirm or reschedule appointments, and view doctor availability in real time, eliminating the need to cross-check separate paper registers. Laboratory technicians will upload diagnostic results such as pathology, X-ray, ECG, and ultrasound reports directly to a patient's electronic record, so that both doctors and patients can retrieve them instantly instead of relying on physical printouts. Billing will be handled within the same platform, automatically calculating charges per visit or service and maintaining a running record of payments, which will support accurate financial reporting. Administrators will oversee the system through a centralized dashboard that consolidates user accounts, department information, appointment volume, and billing summaries, giving them the reporting tools needed to monitor day-to-day operations without manually compiling data from separate sources.

By consolidating these functions into one coherent platform, the proposed system aims to reduce administrative workload, minimize scheduling and record-keeping errors, and give both patients and hospital staff faster, more reliable access to the information they need.

1.2 Selection of Process Model

For the development of the Radium Diagnostic & Consultation Center Hospital Management System (HMS), the **Evolutionary Prototyping Model** has been selected as the software process model.

Explanation of the Model

The Evolutionary Prototyping Model is an iterative process model in which an initial working version (prototype) of the system is built from the best available understanding of requirements. This prototype is then evaluated by stakeholders, refined based on their feedback, and progressively evolved — cycle by cycle — until it matures into the final, accepted system. Unlike the Waterfall Model, it does not assume that requirements are complete and fixed before development starts; instead, it treats early requirements as a starting hypothesis to be tested and corrected through direct stakeholder interaction with a working system.

Evolutionary Prototyping Model — Radium HMS Project

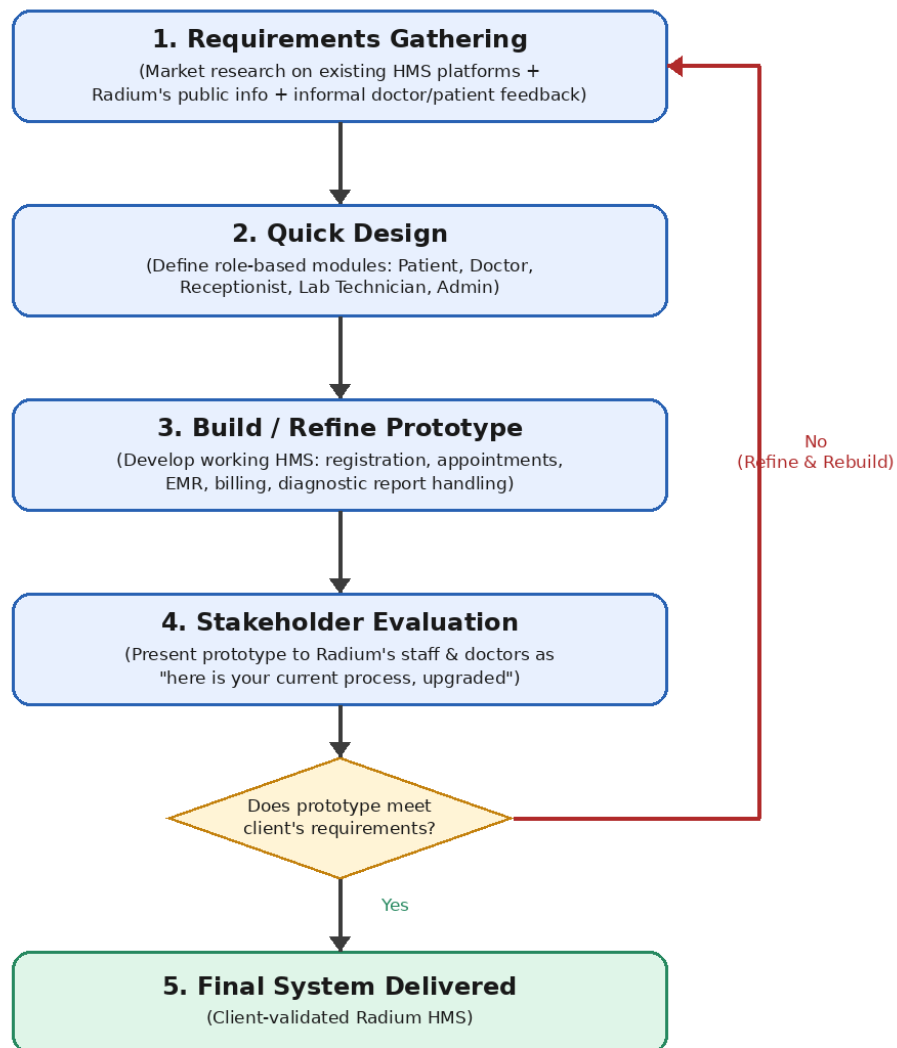


Figure: Evolutionary Prototyping Model applied to the Radium HMS project

Justification for Choosing This Model

This model was chosen because it is the closest match to how the Radium HMS project was actually carried out, and because it directly addresses the main constraint faced during development: the client (Radium Diagnostic & Consultation Center) was not consulted before the first working version of the system was built.

The reasoning behind the selection is as follows:

1. **Requirements were indirect, not client confirmed.** Initial requirements were derived from studying Radium's publicly available information, researching existing/ideal HMS platforms, and gathering informal opinions from a small number of doctors and patients — not from a formal requirements session with hospital management. A model like Waterfall, which assumes complete and signed-off requirements before design begins, would misrepresent this process and offer no way to correct incorrect assumptions later.
2. **A working prototype was built before client validation.** The project's actual sequence — research first, then build a functioning system, then plan to present it to the client — mirrors the prototyping cycle exactly: gather available information → design → build → evaluate with real users.
3. **The client evaluation stage is still pending and expected to drive change.** The upcoming presentation to Radium, framed as "here is your current process, upgraded," is essentially the model's evaluation stage. Since this is the first time the actual client will see the system, feedback at this stage is likely to require revisions to workflows, roles, or features — which the prototyping model explicitly accommodates through its refinement loop, rather than treating such feedback as a failure or scope violation.
4. **The domain benefits from iterative validation.** A Hospital Management System touches sensitive, operationally critical processes (patient records, billing, diagnostic reports), where getting details wrong on the first attempt is likely. Prototyping allows these details — e.g., which departments need report templates, how billing should reconcile across services — to be corrected incrementally with real stakeholder input, rather than discovered too late in a single linear delivery.

2. Timeline Chart / Gantt Chart

Project Timeline – Gantt Chart

Hospital Management System for Radium Diagnostic & Consultation Center

ID	Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
1.	Brainstorm topic & define problem												
2.	Draft background & problem context												
3.	Select & justify process model												
4.	Project Proposal Submission												
5.	SRS: Introduction & Scope/Features												
6.	SRS: Use Case Diagram & Actor Table												
7.	SRS: User Stories & Traceability Matrix												
8.	SRS Submission												
9.	System Design & Trello Board Setup												
10.	UI/UX Prototype in Figma												
11.	Compile Final Report & Gantt Chart												
12.	Final Review & Submission												

AIUB – Software Engineering Course Project | Summer 2025-2026

3. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS)

3.1 Scopes and Features

Scope 1 -- Patient Management

Patient registration, secure login, profile management, and maintenance of personal and contact information. Patients can view their medical history and previous appointments through the system.

Scope 2 -- Appointment & Diagnostic Management

Online appointment booking, rescheduling, and cancellation with real-time doctor availability. Doctors can manage their schedules, and laboratory technicians can upload diagnostic reports, which are accessible to both doctors and patients.

Scope 3 -- Hospital Administration

Management of doctors, departments, billing, payments, user accounts, and administrative reports. Administrators and receptionists can monitor hospital operations, manage records, and generate reports through a centralized dashboard.

3.2 User Story Table

3.2.1 User Story Table

Epic		User stories	Acceptance criteria
Epic 1: Patient Management	User story 1	As a patient, I want to <i>register and log in to the system</i> so that <i>I can access my personal health information securely</i> .	Given a patient provides valid registration details, when they submit the registration form, then an account should be created, and they should be able to log in successfully.
	User story 2	As a patient, I want to <i>manage my profile information</i> so that <i>my contact and personal details stay accurate</i> .	Given a logged-in patient opens their profile, when they update their details and save changes, then the system should store the updated details and confirm the change.
	User story 3	As a patient, I want to <i>view my medical and appointment history</i> so that <i>I can track my past treatments and visits</i> .	Given a logged-in patient with prior records, when they open the medical/appointment history page, then the system should display a chronological list of past visits and treatments.

Epic		User stories	Acceptance criteria
Epic 2: Appointment & Diagnostic Management	User story 1	As a patient, I want to <i>book appointments online</i> so that <i>I do not need to call the hospital</i> .	Given that a patient is logged in and a doctor has available slots, when the patient selects a date and time and confirms, then the appointment should be booked, and confirmation should be shown.
	User story 2	As a doctor, I want to <i>view my daily consultation schedule</i> so that <i>I can prepare for appointments in advance</i> .	Given that a doctor is logged in, when they open their schedule for a selected date, then the system should display all booked appointments for that day in time order.
	User story 3	As a laboratory technician, I want to <i>upload diagnostic reports</i> so that <i>patients and doctors can access them online</i> .	Given a lab technician has completed a test for a patient, when they upload the report and link it to the patient's record, then the report should become viewable to that patient and their doctor.
Epic 3: Hospital Administration	User story 1	As a receptionist, I want to <i>manage appointments and register walk-in patients</i> so that <i>front-desk scheduling becomes easier</i> .	Given a walk-in patient arrives without an online account, when the receptionist registers them and books an appointment, then the patient record and appointment should be created in the system.
	User story 2	As an administrator, I want to <i>manage users and generate reports</i> so that <i>hospital operations can be monitored efficiently</i> .	Given an administrator is logged in, when they generate a report for a selected period, then the system should produce accurate statistics on appointments, revenue, and patients.
	User story 3	As an administrator, I want to <i>manage doctors and departments</i> so that <i>the system reflects the current staff and organizational structure</i> .	Given an administrator needs to update staffing, when they add, edit, or remove a doctor or department, then the change should be reflected across the system immediately.

3.3 Requirements Traceability Matrix

3.3.1 Functional Requirements

FR ID	Requirement	Description	Related User Story
FR-01	Patient Registration	The system shall allow patients to register a new account with their personal and contact details.	Epic 1 – User Story 1
FR-02	Login	The system shall allow all users (patients, doctors, receptionists, lab technicians, administrators) to log in with role-based credentials.	Epic 1 – User Story 1
FR-03	Appointment Booking	The system shall allow patients and receptionists to book, reschedule, or cancel appointments based on doctor availability.	Epic 2 – User Story 1, Epic 3 – User Story 1
FR-04	Doctor Management	The system shall allow administrators to add, update, and remove doctors and assign them to departments.	Epic 3 – User Story 3
FR-05	Diagnostic Report Upload	The system shall allow laboratory technicians to upload diagnostic reports and link them to the correct patient.	Epic 2 – User Story 3
FR-06	Billing	The system shall allow receptionists and administrators to generate bills, calculate charges, and track payment status.	Epic 3 – User Story 1, Epic 3 – User Story 2
FR-07	Report Generation	The system shall allow administrators to generate reports on appointments, revenue, patients, and department performance.	Epic 3 – User Story 2
FR-08	Profile Management	The system allows patients to view and update their profile information.	Epic 1 – User Story 2
FR-09	Medical & Appointment History	The system shall allow patients to view their medical history and past appointment records.	Epic 1 – User Story 3
FR-10	Doctor Schedule	The system allows doctors to view their daily consultation schedule.	Epic 2 – User Story 2

3.3.2 Non-Functional Requirements

NFR ID	Category	Requirement Description
NFR-01	Security	The system should enforce role-based access control and protect sensitive patient data from unauthorized access.
NFR-02	Reliability	The system shall consistently perform its functions (booking, records, billing) without data loss or corruption.
NFR-03	Performance	The system shall respond to user actions such as searching and booking within a few seconds under normal load.
NFR-04	Scalability	The system shall support a growing number of patients, doctors, and diagnostic reports without performance degradation.
NFR-05	Availability	The system shall remain accessible to patients and staff with minimal downtime, including outside office hours for online booking.
NFR-06	Usability	The system should provide a simple, intuitive interface, which is useable by patients and staff with minimal training.
NFR-07	Data Backup	The system shall perform regular automated backups of patient records and diagnostic reports to prevent data loss.

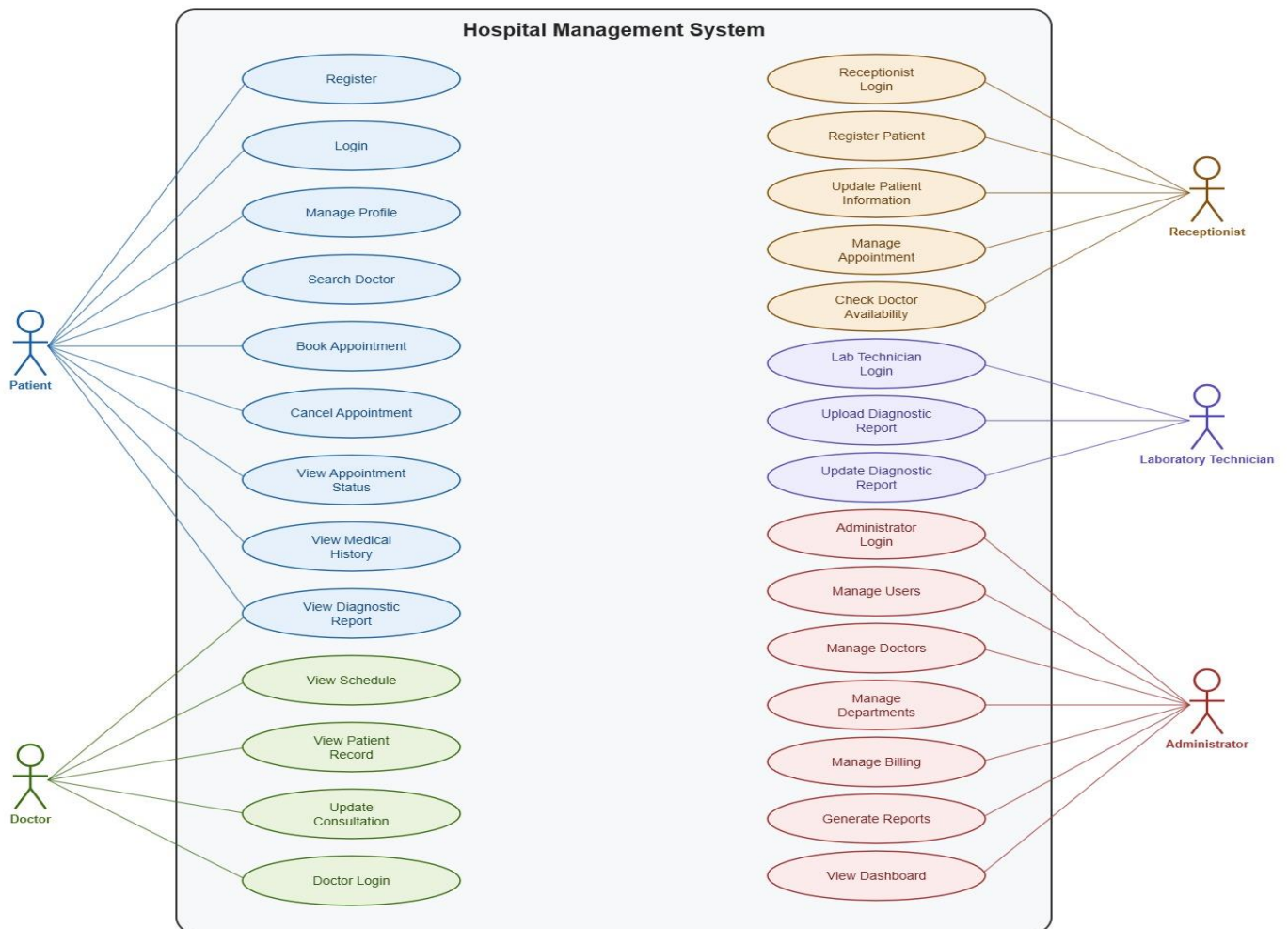
4. SOFTWARE DESIGN

4.1 System Design:

Hospital Management System

Use Case Diagram

Actors: Patient, Doctor, Receptionist, Laboratory Technician, and Administrator, each with their own set of interactions (use cases) within the system. Shared use cases — Login is used by all five actors; View Diagnostic Report is shared between Patient and Doctor; Manage Billing & Payment is shared between Receptionist and Administrator



Patient Role

R

Radium Diagnostic & Consultation Center

Portal Home

Help Desk

Contact

ROLE: VISITOR

Patient Registration Form

Please provide accurate demographic and clinical details to set up your electronic health record.

Full Name

John Doe

Phone Number

+1 (555) 000-0000

Gender

Select Gender

Residential Address

123 Main Street, Suite 4B, New York, NY

Emergency Contact (Name & Phone)

Jane Doe (+1 555-111-2222)

Password

Create robust password

Email Address

john.doe@example.com

Date of Birth

YYYY-MM-DD

Blood Group

e.g. O-Positive

Confirm Password

Repeat password

Register Patient Profile

Back to Login

NAVIGATION

- Dashboard

My Appointments

Medical History

Diagnostic Reports

Profile

Logout

Welcome back, Sarah Jenkins

MRN: RAD-2026-B941 • Welcome to your patient diagnostics console.

Upcoming Appointment

Dr. Aris Vance

Department of Cardiology

CONFIRMED

Monday, Oct 12, 2026

09:30 AM - 10:00 AM

Book New Appointment

Recent Diagnostic Reports

Comprehensive Pathology Screen

READYView Report

Test Date: Oct 02, 2026

Chest X-Ray (Posteroanterior)

READYView Report

Test Date: Sep 28, 2026

Electrocardiogram (ECG)

PENDING

Test Date: Sep 15, 2026

NAVIGATION

- Dashboard

My Appointments

Medical History

Diagnostic Reports

Profile

Logout

Book New Consultation

STEP 1: SELECT DEPARTMENT

Medical Specialty

Cardiology

STEP 2: CHOOSE DOCTOR & SPECIALTY

Dr. Aris Vance

Interventional Cardiologist

Next available: Today at 02:30 PM
Select to View Slots

Dr. Clara Sterling

Cardiovascular Disease Specialist

Next available: Tomorrow at 09:00 AM
Select to View Slots

STEP 3: SELECT DATE

October 2026

S

M

T

W

T

F

S

27

28

29

30

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

STEP 4: SELECT TIME

Morning Slots

09:00 AM

09:30 AM

10:00 AM

11:30 AM

Afternoon Slots

02:00 PM

02:30 PM

03:00 PM

Booking Summary

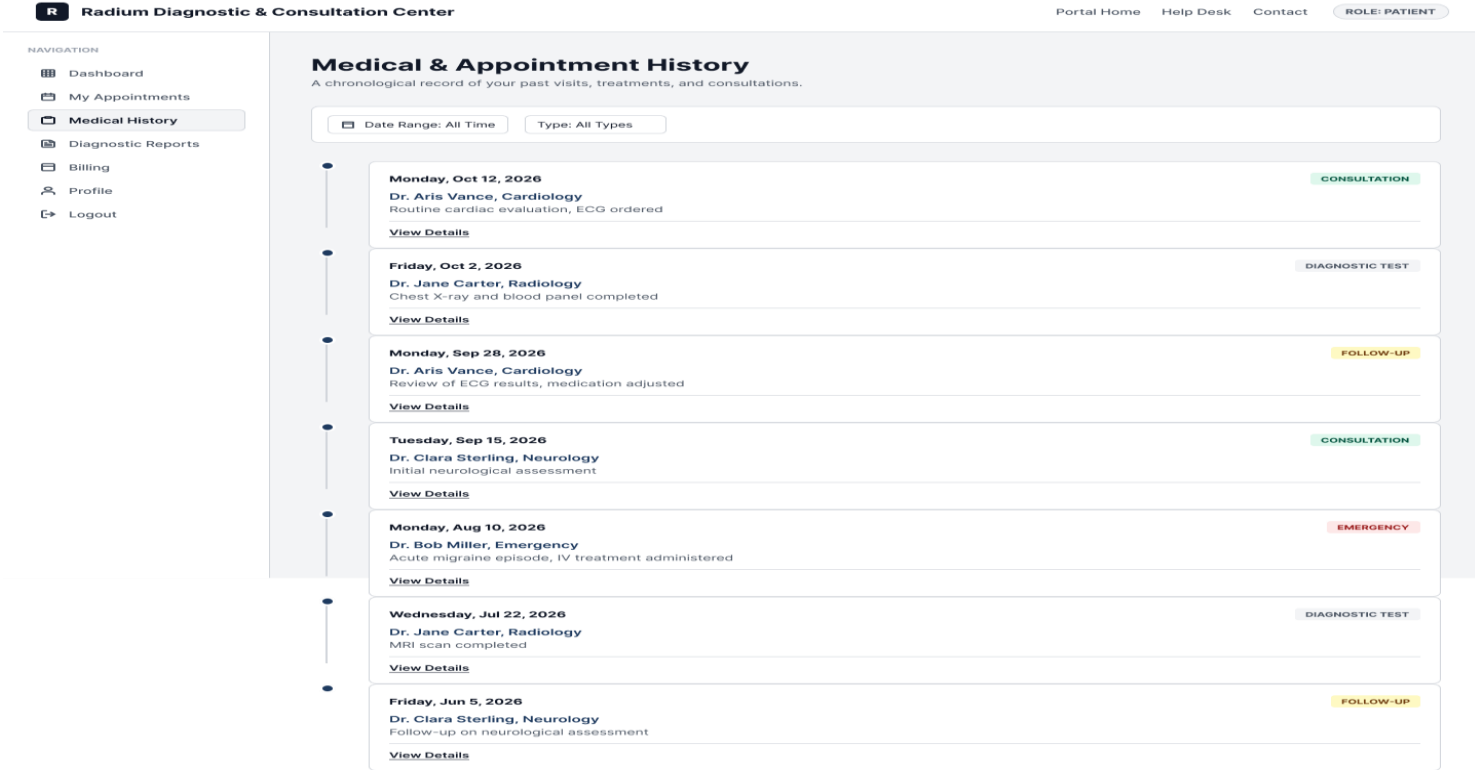
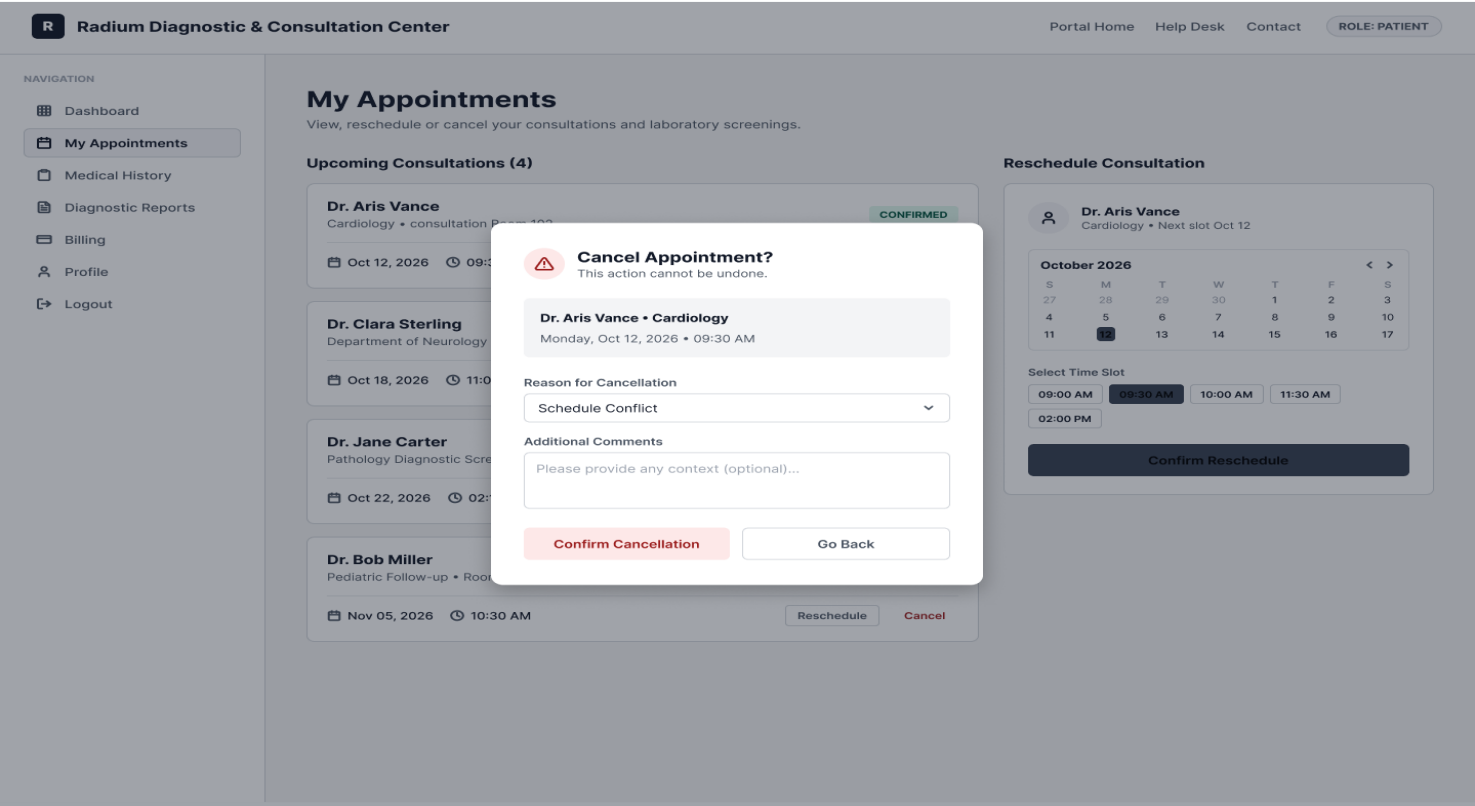
PHYSICIAN
Dr. Aris Vance
Cardiology

DATE & TIME
Monday, Oct 12, 2026
09:30 AM - 10:00 AM

CONSULTATION FEE
\$150.00

Confirm Booking

Cancel



NAVIGATION

- Dashboard
- My Appointments
- Medical History
- Diagnostic Reports**
- Profile
- Logout

READY / FINAL

Report ID: REP-CARD-2026-904

Standard ECG Diagnostic Assessment

Download PDF

Print Report

PATIENT NAME

Sarah Jenkins

MRN (MEDICAL RECORD NUMBER)

RAD-2026-8941

REPORT CATEGORY

Cardiology / Pathology

REFERRING DOCTOR

Dr. Aris Vance

TEST COLLECTION DATE

Oct 02, 2026

AUTHORIZED SIGNATORY

Dr. S. Radium, PhD

Clinical Findings & Technical Assessment

Electrocardiogram Observations:

- Normal sinus rhythm established at a baseline rate of 72 bpm.
- PR Interval: 140ms, QRS Duration: 88ms, QT/QTc: 380/410ms. All intervals measure within standard physiological parameters.
- No diagnostic ST segment elevation, depression, or T-wave inversion indicative of acute ischemia or localized myocardium stress.
- Axis calculation displays standard left-to-center orientation. No hemiblock trends observed.

Conclusion:

Clinically unremarkable cardiogram report. Patient demonstrates standard cardiac electrical conduction properties. Recommend follow-up evaluation as scheduled in the clinical treatment calendar.

NAVIGATION

- Dashboard
- My Appointments
- Medical History
- Diagnostic Reports
- Billing**
- Profile
- Logout

Billing & Payments

View your invoices and manage payments.

Total Outstanding

\$450.00

Last Payment

\$150.00 on Oct 5, 2026

Total Paid (2026)

\$1,200.00

Invoices History

INVOICE #	DATE	DESCRIPT ION	AMOUNT	STATUS	ACTION
RAD-2026-384	Oct 12, 2026	Cardiology consultation	\$150.00	DUE	Pay Now
RAD-2026-340	Oct 05, 2026	Comprehens ive Pathology Screen	\$300.00	DUE	Pay Now
RAD-2026-291	Sep 28, 2026	Chest X-Ray Diagnostic	\$250.00	PAID	Receipt
RAD-2026-212	Sep 15, 2026	Electrocardio gram test fee	\$120.00	PAID	Receipt
RAD-2026-118	Aug 10, 2026	Emergency Ward Admission	\$830.00	OVERDUE	Receipt

Make Payment

Invoice Details

RAD-2026-384 • Cardiology

Cardholder Name

Sarah Jenkins

Card Number

..... 4242

Expiry Date

12 / 29

CVV

...

Payment Amount (\$)

\$150.00

Pay \$150.00

Cancel

NAVIGATION

- Dashboard
- My Appointments
- Medical History
- Diagnostic Reports
- Billing
- Profile**
- Logout

✓ Profile updated successfully.

My Profile

View and update your personal and contact information.

**Sarah Jenkins**

MRN: RAD-2026-8941

Patient since: Jan 2024

ACTIVE ACCOUNT

Personal Information

Full Name *

Sarah Jenkins

Email Address *

sarah.jenkins@email.com

Phone Number *

+1 (555) 234-8901

Date of Birth

March 15, 1992

Gender

Female

Blood Group

A+

Residential Address

142 Maple Street, Apt 4B, Cumilla

Emergency Contact (Relationship) *

Michael Jenkins — (555) 876-5432

Change Password

Current Password

New Password

Confirm New Password

Save Changes

Cancel

Role Doctor

R

Radium Diagnostic

HOSPITAL MANAGEMENT SYSTEM

Select Portal Role

Doctor

Email Address

e.g. Doctor@example.com

Password

Sign In to Portal

- NAVIGATION

- ☒ Dashboard
- ☐ My Schedule
- ☐ Appointments
- ☐ Patient Records
- ☐ Diagnostic Reports
- ☐ Profile
- ☐ Logout

Dr. Aris Vance

Department of Cardiology • Monday, Oct 12, 2026

 Change Date

TOTAL APPOINTMENTS TODAY

14

COMPLETED

8

PENDING

4

CANCELLED

2

Today's Consultations List

Time Slot	Patient Name	Type	Status	Actions
09:00 AM	Sarah Jenkins	First Consultation	Completed	Manage
09:30 AM	Marcus Brody	Follow-up	Completed	Manage
10:00 AM	Timothy Vance	Report Review	Confirmed	Manage
11:00 AM	Elena Rostova	First Consultation	Pending	Manage

- ☐ Dashboard
- ☒ **My Schedule**
- ☐ Appointments
- ☐ Patient Records
- ☐ Diagnostic Reports
- ☐ Profile
- ☐ Logout

Set your weekly consultation hours and block off unavailable slots.

32 hrs

TIME	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
09:00 AM			Booked				
09:30 AM	Booked						
10:00 AM	Booked						
10:30 AM							
11:00 AM		Booked					
11:30 AM					Booked		
12:00 PM							
12:30 PM							
01:00 PM							
01:30 PM							
02:00 PM							
02:30 PM				Booked			
03:00 PM				Booked			
03:30 PM							
04:00 PM							
04:30 PM							

Reset to Default

- Dashboard
- My Schedule
- **Appointments**
- Patient Records
- Diagnostic Reports
- Profile
- Logout

View and manage your upcoming and past patient appointments.

Status: All

Patient Name	Date	Time	Type	Status	Action
Rahim Uddin	2025-07-15	10:00 AM	Consultation	Confirmed	View
Fatema Begum	2025-07-15	10:30 AM	Follow-up	Confirmed	View
Kamal Hossain	2025-07-15	11:00 AM	Consultation	Pending	View
Nasreen Akter	2025-07-16	09:00 AM	Consultation	Confirmed	View
Jamal Mia	2025-07-16	09:30 AM	Follow-up	Cancelled	View
Salma Khatun	2025-07-16	10:00 AM	Consultation	Pending	View
Abdul Karim	2025-07-17	11:30 AM	Consultation	Confirmed	View

Previous 1 2 3 Next

NAVIGATION

- Dashboard
- My Schedule
- Appointments
- Patient Records**
- Diagnostic Reports
- Profile
- Logout

Patient Records

Search and review medical records for patients under your care.

Search

PATIENT NAME	MRN	LAST VISIT DATE	DEPARTMENT	ACTION
Sarah Jenkins	RAD-2026-8941	Oct 12, 2026	Cardiology	View Full Record
Marcus Brody	RAD-2026-7823	Oct 10, 2026	Radiology	View Full Record
Timothy Vance	RAD-2026-6547	Oct 8, 2026	Neurology	View Full Record
Elena Rostova	RAD-2026-5129	Oct 5, 2026	Orthopedics	View Full Record
Robert Downey	RAD-2026-4382	Sep 22, 2026	Neurology	View Full Record

Showing 1 to 5 of 5 patient records

Previous123Next

NAVIGATION

- Dashboard
- My Schedule
- Appointments
- Patient Records
- Diagnostic Reports**
- Profile
- Logout

Diagnostic Reports

Review lab and imaging results for your patients.

Type: All Report Types

PATIENT NAME	REPORT TYPE	TEST DATE	STATUS	ACTION
Sarah Jenkins	Standard ECG	Oct 12, 2026	COMPLETED	View Report
Marcus Brody	Blood Panel	Oct 10, 2026	COMPLETED	View Report
Timothy Vance	Chest X-Ray	Oct 08, 2026	PENDING REVIEW	View Report
Elena Rostova	MRI Brain	Oct 05, 2026	COMPLETED	View Report
John Constantine	Liver Function Test	Oct 03, 2026	PENDING REVIEW	View Report
Clara Oswald	Thyroid Panel	Sep 28, 2026	COMPLETED	View Report
David Tennant	Cardiac Stress Test	Sep 25, 2026	IN PROGRESS	View Report

Showing 1 to 7 of 24 patient reports

Previous123Next

NAVIGATION

- ☐ Dashboard
- ☐ My Schedule
- ☐ Patient Records
- ☐ Diagnostic Reports
- ☒ Profile
- ☐ Logout

✔ Profile updated successfully.

My Profile

View and update your personal and professional information.



Dr. Aris Vance
DID: RAD-DOC-1042
Department of Cardiology

ACTIVE ACCOUNT

Personal & Professional Information

Full Name *

Dr. Aris Vance

Email Address *

aris.vance@radlum.com

Phone Number *

+1 (555) 739-2210

Date of Birth

June 22, 1978

Gender

Male

Blood Group

O+

Specialization *

Cardiology

Qualification

MBBS, MD (Cardiology)

License Number

MED-BD-20145

Department

Cardiology

Residential Address

58 Riverside Drive, Daudkandi, Cumilla

Consultation Fee

1,500 BDT

Change Password

Current Password

New Password

Confirm New Password

Save Changes

Cancel

Role Administrator

R

Radium Diagnostic

HOSPITAL MANAGEMENT SYSTEM

Select Portal Role

Admin

▼

Email Address

e.g. admin@example.com

Password

Sign In to Portal

System Administration Console

Overview of diagnostic center performance, departments, and user accounts.

TOTAL PATIENTS
3,481

TOTAL DOCTORS
24

TODAY'S APPOINTMENTS

58

REVENUE THIS MONTH

\$42,190

Recent Scheduled Appointments

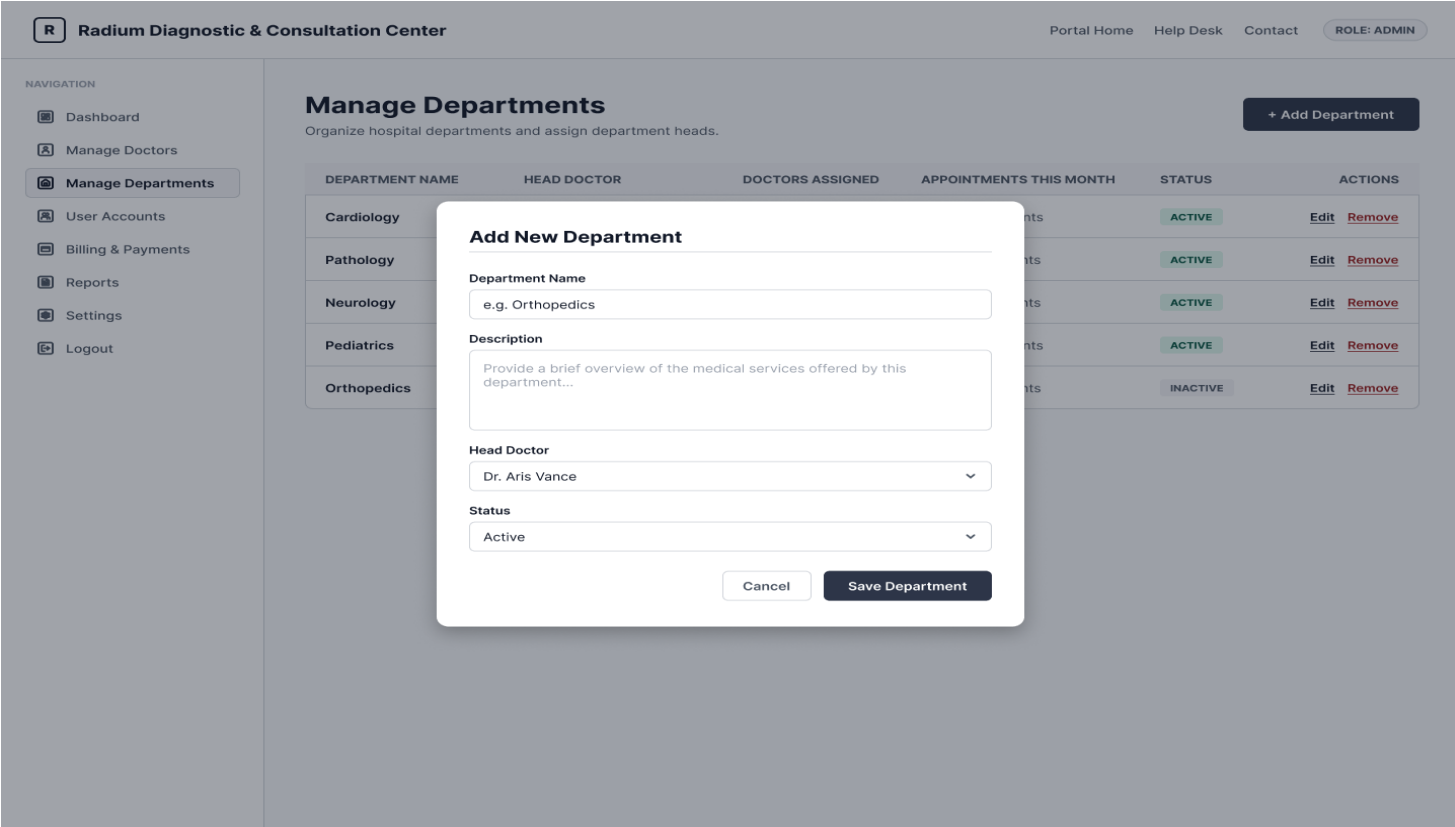
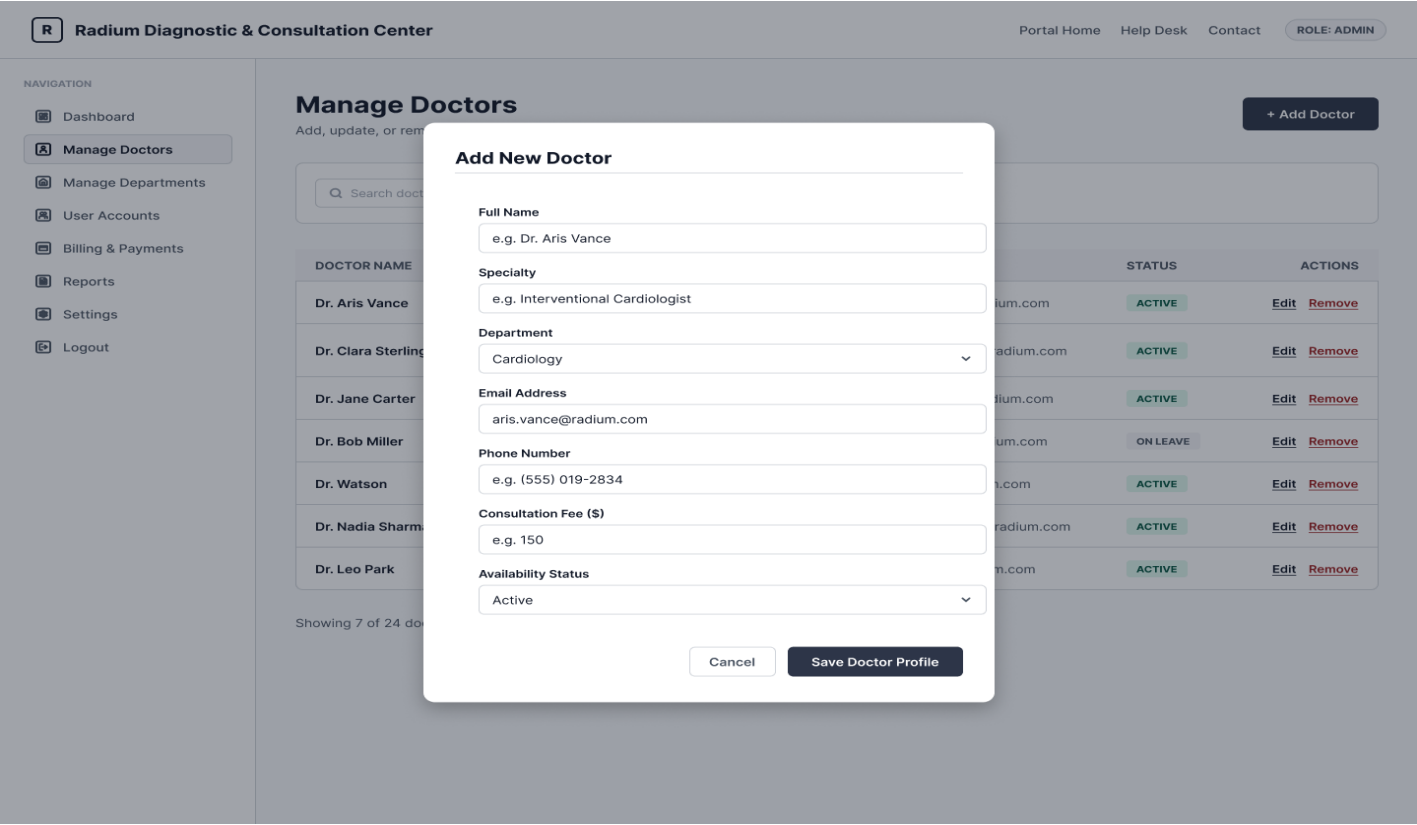
Patient Name	Doctor	Department	Date	Status
Robert Downey	Dr. Vance	Cardiology	Oct 12, 2026	Confirmed
Clara Oswald	Dr. Carter	Pathology	Oct 12, 2026	Pending
Bruce Banner	Dr. Vance	Cardiology	Oct 11, 2026	Completed

Quick Actions

+ Add Doctor Profile

+ Add Department

Generate System Report



R

Radium Diagnostic & Consultation Center

Portal Home

Help Desk

Contact

ROLE: ADMIN

NAVIGATION

Dashboard

Manage Doctors

Manage Departments

User Accounts

Billing & Payments

Reports

Settings

Logout

User Accounts

Manage login access and roles for all system users.

+ Add User

Q Search by name or email...

All Roles

All Statuses

NAME

Sarah Jenkins

Dr. Aris Vance

Dr. Jane Carter

Michael Torres

Priya Nair

James Wilson

Emily Chen

Admin User

Full Name

e.g. Sarah Jenkins

Email Address

sarah.jenkins@email.com

Role

Patient

Temporary Password

tEmp_pAss982!

Generate

☒ Send account invitation email

Cancel

Create Account

ED

24

ACTIVE

Edit

Deactivate

23

ACTIVE

Edit

Deactivate

23

ACTIVE

Edit

Deactivate

23

ACTIVE

Edit

Deactivate

23

ACTIVE

Edit

Deactivate

24

ACTIVE

Edit

Deactivate

24

DEACTIVATED

Edit

Reactivate

3

ACTIVE

Edit

Deactivate

Showing 1 to 8 of 42 entries

Previous

1

2

3

Next

NAVIGATION

Dashboard

Manage Doctors

Manage Departments

User Accounts

Billing & Payments

Reports

Settings

Logout

Billing & Payments Overview

Monitor hospital-wide revenue, outstanding invoices, and payment activity.

TOTAL REVENUE THIS MONTH

\$42,190

▲ 12%

TOTAL OUTSTANDING

\$12,480

OVERDUE INVOICES

8

INVOICES ISSUED TODAY

14

Date Range: Last 30 Days

Status: All Statuses

Q Search patient name or invoice #...

Filter

INVOICE #

PATIENT

DEPARTMENT

AMOUNT

DATE ISSUED

STATUS

ACTION

INV-2026-042

Sarah Jenkins

Cardiology

\$1,250.00

Oct 12, 2026

PAID

View

INV-2026-041

Marcus Brody

Radiology

\$890.00

Oct 10, 2026

PENDING

View

INV-2026-040

Timothy Vance

Neurology

\$2,100.00

Oct 8, 2026

OVERDUE

View

INV-2026-039

Elena Rostova

Orthopedics

\$1,750.00

Oct 5, 2026

PAID

View

INV-2026-038

John Constantine

Cardiology

\$950.00

Oct 3, 2026

PAID

View

INV-2026-037

Clara Oswald

Radiology

\$1,100.00

Sep 28, 2026

PENDING

View

INV-2026-036

David Tennant

Emergency

\$3,200.00

Sep 25, 2026

OVERDUE

View

INV-2026-035

Robert Downey

Neurology

\$780.00

Sep 22, 2026

PAID

View

Showing 1 to 8 of 24 results

Previous

1

2

3

Next

R

Radium Diagnostic & Consultation Center

Portal Home

Help Desk

Contact

ROLE: ADMIN

Page 25 of 34

NAVIGATION

- Dashboard
- Manage Doctors
- Manage Departments
- User Accounts
- Billing & Payments
- Reports
- Settings
- Logout

System Reports

Generated report for selected date range.

From: Oct 01, 2026

To: Oct 31, 2026

Comprehensive Summary

Generate Report

TOTAL APPOINTMENTS

1,247

TOTAL REVENUE

\$186,450

NEW PATIENTS

342

AVERAGE WAIT TIME

14 min

Appointments by Department

Cardiology	342
Pathology	412
Neurology	189
Orthopedics	100
Pediatrics	204

Revenue Trend (Jul - Dec 2026)



Department Performance

DEPARTMENT	APPOINTMENTS	REVENUE	AVG. RATING	TOP DOCTOR
Cardiology	342	\$51,300	4.8	Dr. Vance
Pathology	412	\$61,800	4.7	Dr. Carter
Neurology	189	\$28,350	4.9	Dr. Sterling
Pediatrics	204	\$30,600	4.6	Dr. Miller
Orthopedics	100	\$14,400	4.5	Dr. Watson

Export as PDF

Export as CSV

NAVIGATION

- Dashboard
- Manage Doctors
- Manage Departments
- User Accounts
- Billing & Payments
- Report Generation
- Settings
- Logout

System Settings

Configure general hospital information and system preferences.

Hospital Information

Hospital Name	Email Address
Radium Diagnostic & Consultation Center	admin@radiumcenter.com
Phone Number	Operating Hours
+880 1234-567890	8:00 AM - 10:00 PM
Address	
Daudkandi, Cumilla	

Notification Settings

Appointment Reminders	☒
SMS Notifications	☒
Email Notifications	☒

Security Settings

Two-Factor Authentication (2FA)	Session Timeout
Require 2FA for all administrative accounts ☒	30 minutes

Backup & Data

Backup Frequency	Last automated backup: Aug 24, 2026 at 2:00 AM
Daily	<button>Run Backup Now</button>

Save All Settings

Cancel

Role Receptionist

R

Radium Diagnostic & Consultation Center

Portal Home

Help Desk

Contact

ROLE: VISITOR

R

Radium Diagnostic

HOSPITAL MANAGEMENT SYSTEM

Select Portal Role

Receptionist

Email Address

e.g. Receptionist@example.com

Password

Sign In to Portal

R

Radium Diagnostic & Consultation Center

Portal Home

Help Desk

Contact

ROLE: RECEPTIONIST

NAVIGATION

Dashboard

Register Patient

Appointments

Doctor Schedule

Billing

Profile

Logout

Receptionist Care Desk

Manage patient admissions, walk-in consultation queues, and appointments.

Today's Patient Schedule

TIME	PATIENT	DOCTOR	DEPT	STATUS	ACTIONS
09:30 AM	Sarah Jenkins	Dr. Vance	Cardiology	CONFIRMED	Confirm
10:15 AM	David Tennant	Dr. Carter	Pathology	PENDING	Confirm
11:00 AM	Martha Jones	Dr. Vance	Cardiology	CONFIRMED	Confirm

Quick Register Walk-in

Patient Full Name

e.g. John Doe

Phone Number

e.g. +1 555-0199

Department

Select Specialty

Consulting Doctor

Assign Doctor

Preferred Slot

Select Time Slot

Register & Admit Patient

Doctor Availability status

Dr. Aris Vance

Cardiology

IN CONSULTATION

Dr. Jane Carter

Pathology

AVAILABLE

Dr. Bob Miller

Pediatrics

OFF DUTY

NAVIGATION

- Dashboard
- Register Patient
- Appointments
- Doctor Schedule
- Billing
- Profile
- Logout

Appointment Management

View, confirm, reschedule, or cancel patient appointments.

Oct 12, 2026 - Oct 18, 2026

All Doctors

All Statuses

Q

Search patient name...

Search

TIME	PATIENT NAME	DOCTOR	DEPARTMENT	STATUS	ACTIONS
09:30 AM	Sarah Jenkins	Dr. Vance	Cardiology	CONFIRMED	Reschedule Cancel
10:15 AM	David Tennant	Dr. Carter	Pathology	PENDING	Confirm Reschedule Cancel
11:00 AM	Martha Jones	Dr. Vance	Cardiology	CONFIRMED	Reschedule Cancel
11:45 AM	Donna Noble	Dr. Miller	Pediatrics	CANCELLED	Reschedule
02:00 PM	Rose Tyler	Dr. Sterling	Neurology	RESCHEDULED	Reschedule Cancel
03:30 PM	Amy Pond	Dr. Carter	Pathology	PENDING	Confirm Reschedule Cancel

Showing 1 to 6 of 28 entries

Previous

1

2

3

Next

NAVIGATION

- Dashboard
- Register Patient
- Appointments
- Doctor Schedule
- Billing
- Profile
- Logout

Billing & Invoice Management

Create comprehensive diagnostic invoices and view recent payment status.

Create New Invoice

Select Patient

Sarah Jenkins

Q

Select Appointment

Oct 12, 2026 — Dr. Vance, Cardiology

Consultation Fee (\$)

\$150.00

Additional Line Items

ITEM DESCRIPTION	QTY	PRICE	TOTAL
Blood Panel Test	1	\$85.00	\$85.00
ECG Test	1	\$120.00	\$120.00
+ Add Line Item			

Subtotal

\$355.00

Tax (5%)

\$17.75

Grand Total

\$372.75

Generate Invoice

Recent Invoices

INVOICE #	PATIENT	AMOUNT	STATUS
INV-2026-042	John Watson Oct 12, 2026	\$355.00	PAID
INV-2026-041	Sarah Jenkins Oct 12, 2026	\$235.00	DUE
INV-2026-040	Clara Oswald Oct 11, 2026	\$150.00	PAID
INV-2026-039	Bruce Banner Oct 10, 2026	\$420.00	OVERDUE
INV-2026-038	Peter Parker Oct 10, 2026	\$150.00	PAID

NAVIGATION

- Dashboard
- Register Patient
- Appointments
- Doctor Schedule
- Billing
- Profile
- Logout

Patient Registration

Register a new patient and create their electronic health record.

Full Name

e.g. John Doe

Email Address

e.g. john.doe@example.com

Phone Number

e.g. +1 (555) 019-2834

Date of Birth

YYYY-MM-DD

Gender

Select Gender

Blood Group

Select Blood Group

Residential Address

Street Address, City, State, ZIP Code

Emergency Contact Name

e.g. Jane Doe

Emergency Contact Phone

e.g. +1 (555) 019-5678

Password

Create robust password

Confirm Password

Repeat created password

Register Patient Profile

NAVIGATION

- Dashboard
- Register Patient
- Appointments
- Doctor Schedule
- Billing
- Profile
- Logout

Doctor Schedule

View doctor availability and consultation schedules.

Consultation Date

Oct 14, 2026

Department

All Departments

Q Search doctor name...

Search

DOCTOR NAME	DEPARTMENT	AVAILABLE SLOTS	STATUS	ACTIONS
Dr. Aris Vance	Cardiology	09:00 AM – 12:00 PM, 02:00 PM – 05:00 PM	AVAILABLE	View Schedule
Dr. Jane Carter	Pathology	10:00 AM – 01:00 PM	IN CONSULTATION	View Schedule
Dr. Bob Miller	Pediatrics	09:00 AM – 11:00 AM, 03:00 PM – 05:00 PM	AVAILABLE	View Schedule
Dr. Clara Sterling	Neurology	—	OFF DUTY	View Schedule
Dr. Watson	Orthopedics	11:00 AM – 01:00 PM, 04:00 PM – 06:00 PM	AVAILABLE	View Schedule

NAVIGATION

- Dashboard
- Register Patient
- Appointments
- Doctor Schedule
- Billing
- Profile
- Logout

My Profile

View and update your account information.



Rina Khatun
Front Desk Receptionist
Employee ID: EMP-2026-0045

Personal Details

Full Name

Rina Khatun

Email Address

rina.khatun@radium.com

Phone Number

+1 (555) 019-8833

Department (Read-only)

Front Desk

Date of Joining (Read-only)

Jan 15, 2024

Change Password

Save Changes

Role Lab-Technician



Radium Diagnostic
HOSPITAL MANAGEMENT SYSTEM

Select Portal Role

Lab-Technician

Email Address

e.g. Lab-Technician@example.com

Password

Sign In to Portal

- NAVIGATION
- Dashboard

Upload Report

Patient Reports

Profile

Logout

Laboratory Consultation Console

Upload diagnostic results and track recent digital lab records.

Upload New Diagnostic Report

Patient ID / Name Search

Report Type

Test Date

Findings Summary

Click to upload report files

Submit & Publish Diagnostic Report

Recent Uploaded Results

PATIENT NAME	TYPE	STATUS
John Constantine	Pathology	READY
Clara Oswald	X-Ray	READY
David Tennant	ECG	PENDING

- NAVIGATION
- Dashboard

Upload Report

Patient Reports

Profile

Logout

Patient Diagnostic Reports

Browse, search, and manage all diagnostic reports.

Search by patient name/ID...

Report Type

Status

Search

REPORT ID	PATIENT NAME	TEST TYPE	TEST DATE	STATUS	ACTIONS
REP-2026-904	Sarah Jenkins	ECG	Oct 02, 2026	READY	View Edit
REP-2026-903	John Constantine	Pathology	Oct 01, 2026	READY	View Edit
REP-2026-902	Clara Oswald	X-Ray	Sep 30, 2026	READY	View Edit
REP-2026-901	David Tennant	ECG	Sep 29, 2026	PENDING	View Edit
REP-2026-900	Amy Pond	Blood Panel	Sep 28, 2026	DRAFT	View Edit

Showing 1 to 5 of 42 entries

Previous

1

2

3

Next

NAVIGATION

- Dashboard
- Upload Report**
- Patient Reports
- Profile
- Logout

Upload Diagnostic Report

Submit new diagnostic test results and link them to patient records.

Diagnostic Details

Patient ID / Name Search

e.g. RAD-2026-8941

Report Type

Select Diagnostics Type

Test Date

YYYY-MM-DD

Findings Summary

Enter diagnostic observations...



Click to upload report files

PDF, PNG or JPG up to 10MB

Submit & Publish Diagnostic Report

Recent Uploaded Results

PATIENT NAME / ID	TYPE	STATUS
Sarah Jenkins REP-2026-904	ECG	READY
John Constantine REP-2026-903	Pathology	READY
Clara Oswald REP-2026-902	X-Ray	PENDING
David Tennant REP-2026-901	ECG	DRAFT

NAVIGATION

- Dashboard
- Upload Report
- Patient Reports
- Profile**
- Logout

My Profile

View and update your account information.

AH

Arif HossainSenior Lab Technician
Employee ID: EMP-2026-0078

Personal Details

Full Name

Arif Hossain

Email Address

arif.hossain@radium.com

Phone Number

+1 (555) 019-3322

Department (Read-only)

Diagnostics Lab

Date of Joining (Read-only)

Mar 10, 2023

Change Password

Save Changes

5. CONCLUSION

The Radium Diagnostic & Consultation Centre currently relies on manual, phone-based processes for appointment booking, patient record keeping, diagnostic report handling, and billing, which creates inefficiencies as patient volume grows and increases the risk of scheduling conflicts, lost records, and delayed care. The proposed Hospital Management System addresses these gaps by consolidating patient registration, appointment scheduling, doctor and department management, electronic medical records, diagnostic report handling, and billing into a single, role-based platform accessible to patients, doctors, receptionists, laboratory technicians, and administrators.

Given that the initial requirements were assembled from research into existing HMS platforms, publicly available information about Radium's operations, and informal feedback from doctors and patients rather than a formal requirements session with the client, the Evolutionary Prototyping Model was adopted as the guiding process model. This approach allowed a working prototype to be developed early while explicitly leaving room for the client's evaluation and feedback to refine the system further, ensuring that the final delivered platform reflects Radium's actual operational needs rather than assumptions alone.

Overall, the proposed system lays a strong foundation for modernizing Radium Diagnostic & Consultation Centre's operations, and the next phase of development will focus on incorporating client feedback to finalize workflows, roles, and reporting requirements before full deployment.

6. Appendix

Appendix A: Group Trello Board for Project Task Progression

For effective project management, task distribution, and progress monitoring, the project team used a Trello board titled “**Radium Diagnostic Center (Group C)**”. The Trello board was used to divide the project tasks among the group members and monitor the progression of each assigned responsibility. Each team member was assigned specific tasks based on different modules of the **Radium Diagnostic Center Management System**.

A.1 Task Distribution Among Group Members

Project responsibilities were divided among the three group members according to distinct functional modules of the Radium Diagnostic Center Hospital Management System, ensuring that each member owned both the requirements documentation and the corresponding UI/UX design for their assigned module.

Trello Board Link: <https://trello.com/b/UB4WUSAL>

Group Member	Assigned Module	Key Responsibilities
Samia Afruz Asha (24-57314-2)	Patient Module	• Drafted Patient user stories • Documented Patient functional requirements (FR-08, FR-09) • Completed RTM entries for Patient rows • Prepared use case write-up for the Patient actor • Authored the Process Model section • Designed the Patient role UI/UX in Figma
Munish Sarker (24-57375-2)	Doctor & Admin Module	• Drafted Admin user stories • Documented Doctor and Admin functional requirements (FR-04, FR-07, FR-10) • Completed RTM entries for Doctor and Admin rows • Authored the System Design section • Wrote the Introduction and Background section • Designed the Doctor and Admin role UI/UX in Figma
Rinko Rani Vadra (24-58043-2)	Reception & Laboratory Module	• Drafted Reception and Laboratory user stories • Documented Reception/Laboratory functional requirements (FR-02, FR-03, FR-05, FR-06) • Completed RTM entries for Reception and Laboratory rows • Prepared the Project Timeline and Gantt Chart • Designed the Reception and Laboratory role UI/UX in Figma